

Sources and Nature of Data

1. Sources of Data

A **data source** is the place or method from which data is collected.

A. Primary Data

Data collected **directly by the researcher or organization** for a specific purpose.

Examples:

- Surveys and questionnaires
- Interviews
- Observations
- Experiments
- Sensors and IoT devices
- Customer feedback

Example: A college conducts a survey to find students' preferred programming languages.

B. Secondary Data

Data that has been **already collected by someone else** and is reused for another purpose.

Examples:

- Government reports
- Research papers
- Websites
- Books and journals
- Company databases
- Public datasets
- Historical records

Example: Using Census data published by the Government of India for research.

2. Nature of Data

Data can be classified according to its **format and organization**. The three major types are:

1. **Structured Data**
2. **Semi-Structured Data**
3. **Unstructured Data**

Comparison

Type	Organization	Examples	Common Storage
Structured	Highly organized, fixed schema	SQL tables, student records	RDBMS, spreadsheets
Semi-structured	Partially organized, flexible schema	JSON, XML, emails	MongoDB, document stores
Unstructured	No predefined structure	Images, videos, audio, PDFs	File systems, object storage

3. Structured Data

Structured data is data that follows a **fixed format or predefined schema**. It is usually arranged in **rows and columns**.

Example:

Student_ID	Name	Branch	Marks
101	Ravi	Computer	85
102	Amit	Computer	78

Characteristics

- Organized in rows and columns
- Has a predefined schema
- Easy to search and analyze
- Usually stored in relational databases
- Can be queried using **SQL**

Examples: Student databases, bank transactions, employee records, inventory data.

4. Semi-Structured Data

Semi-structured data does not follow a strict tabular structure but contains **tags, keys, attributes, or metadata** that provide some organization.

Example: JSON

```
{
  "name": "Ravi",
  "branch": "Computer",
  "marks": 85
}
```

Characteristics

- No fixed table structure

- Contains keys, tags, or attributes
- More flexible than structured data
- Easier to modify or expand
- Common in web applications and APIs

Examples: JSON, XML, HTML, emails, log files.

5. Unstructured Data

Unstructured data is data that does **not have a predefined data model or fixed structure**.

Examples

- Images
- Videos
- Audio files
- Social media posts
- Documents
- Presentations
- Scanned PDFs

Characteristics

- No predefined schema
 - Difficult to store and analyze using traditional relational databases
 - Often requires **AI, Machine Learning, NLP, or Computer Vision** for analysis
 - Represents a large portion of modern data
-

Exam-Oriented Short Answer

Structured data: Data organized in a fixed schema, generally in rows and columns.

Example: SQL database.

Semi-structured data: Data without a rigid tabular structure but containing tags or keys.

Example: JSON and XML.

Unstructured data: Data without a predefined structure. Example: images, videos, audio and documents.

Key point to remember:

Structured → **Table** | **Semi-structured** → **Tags/Keys** | **Unstructured** → **No fixed structure**